Number — Negative integers · Prime factors · Indices · Priority of operations

No calculator. Show your working. ★ = challenge question.

A · NEGATIVE INTEGERS

1 The temperature in Oslo was -6 °C at 7 am and 3 °C at 2 pm.

a Work out the rise in temperature between 7 am and 2 pm.

b By midnight the temperature had fallen by 11 °C compared with 2 pm. What was the temperature at midnight?

c Work out $-3-7$.

> **a** From -6 up to 3: $3-(-6)=3+6=9$ °C.
> **b** $3-11=-8$ °C.
> **c** Start at -3, go 7 further down: -10.

2 Work out

a -4×6 **b** $(-3)\times(-7)$ **c** $45\div(-9)$ **d** $-36\div(-4)$

e $-2\times(-3)\times(-5)$

> Same signs → positive, different signs → negative. **a** -24 **b** 21 **c** -5 **d** 9 **e** $(-2)(-3)=6$, then $6\times(-5)=-30$ (three negatives → negative).

3 Substitute the values into each formula and work out the answer.

a $v=u+at$ when $u=-5$, $a=3$ and $t=4$

b $A=2b-c^2$ when $b=-3$ and $c=-4$

> **a** $v=-5+3\times 4=-5+12=7$.
> **b** $c^2=(-4)^2=16$, so $A=2\times(-3)-16=-6-16=-22$.

4 Maya says that $(-5)^2=-25$. Explain her mistake and write the correct value.

> Squaring means multiplying the number by itself: $(-5)^2=(-5)\times(-5)=+25$ — two negatives give a positive. -25 would be $-5^2=-(5\times 5)$, where only the 5 is squared.

B · PRIME FACTOR DECOMPOSITION, HCF AND LCM

5 Write each number as a product of its prime factors, using index notation.

a 84 **b** 360

> **a** $84=2\times 42=2\times 2\times 21=2\times 2\times 3\times 7=2^2\times 3\times 7$.
> **b** $360=2\times 180=2\times 2\times 90=2\times 2\times 2\times 45=2^3\times 3^2\times 5$.

6 Use your answers to question 5 to find

a the HCF of 84 and 360 **b** the LCM of 84 and 360

> $84 = 2^2 \times 3 \times 7$ and $360 = 2^3 \times 3^2 \times 5$.
>
> **a** HCF = the primes they share, lowest powers: $2^2 \times 3 = 12$.
>
> **b** LCM = every prime that appears, highest powers: $2^3 \times 3^2 \times 5 \times 7 = 2520$.

7 Two buses leave a station at 9:00. Bus A leaves every 12 minutes and bus B every 18 minutes. At what time do they next leave together?

> They meet at multiples of both 12 and 18, i.e. the LCM: $12 = 2^2 \times 3$, $18 = 2 \times 3^2$, LCM $= 2^2 \times 3^2 = 36$ minutes. Next together at 9:36.

C · INDICES

8 Write each calculation as a single power.

a $7^4 \times 7^3$ **b** $5^9 \div 5^4$ **c** $(2^3)^4$ **d** $3^6 \times 3 \div 3^2$ **e** $10^8 \div 10^8$

> Multiply → add the indices, divide → subtract, power of a power → multiply.
>
> **a** 7^7 **b** 5^5 **c** 2^{12} **d** $3^{6+1-2} = 3^5$ **e** $10^0 = 1$ (any number to the power 0 is 1).

9 Evaluate $\dfrac{2^5 \times 2^3}{2^6}$

> $\dfrac{2^8}{2^6} = 2^{8-6} = 2^2 = 4$.

10 Which calculation does **not** give the same answer as the others? Explain.

A $(2 \times 3)^2$ **B** 2×3^2 **C** $2^2 \times 3^2$ **D** 6^2

> A $= 6^2 = 36$, C $= 4 \times 9 = 36$, D $= 36$. B $= 2 \times 9 = 18$ — the square applies only to the 3, because indices are done before multiplication. **B**.

D · PRIORITY OF OPERATIONS

11 Work out

a $4 + 3 \times 2^2$ **b** $(4 + 3) \times 2^2$ **c** $20 - \sqrt{36} \div 2$ **d** $(10 - 2^3)^2 \times 3$

e $\sqrt[3]{27} + 5 \times 2$ **f** $100 \div (3 + 2)^2$

> **a** $4 + 3 \times 4 = 16$ **b** $7 \times 4 = 28$ **c** $20 - 6 \div 2 = 20 - 3 = 17$ **d** $(10 - 8)^2 \times 3 = 4 \times 3 = 12$ **e** $3 + 10 = 13$ **f** $100 \div 25 = 4$.

12 Insert one pair of brackets to make this calculation correct: $3 + 5 \times 2 - 1 = 15$

> Without brackets: $3 + 10 - 1 = 12$. Forcing the addition first: $(3 + 5) \times 2 - 1 = 16 - 1 = 15$ ✓.

13 Evaluate $\dfrac{5 \times 2^4}{4 \times 10}$ and $\dfrac{(7-4)^2}{3^2-6}$

> $\dfrac{5 \times 16}{40} = \dfrac{80}{40} = 2$. $\dfrac{3^2}{9-6} = \dfrac{9}{3} = 3$. (The fraction line acts like brackets: work out the top and the bottom first.)

Check: a sign for every negative answer · index laws only work for the *same* base · brackets → indices → ÷ × → + −.